

Course Assignment Brief

Course Title: Cisco Network Security

Assignment Title: Design and Implementation of a Secure Enterprise Network

Assignment Type: Group Project (Composite Assignment)

Weighting: 20% of Final Grade

Duration: 4 Weeks

1. Introduction

This assignment requires students to design, implement, and document a secure small enterprise network using Cisco networking and security technologies. The project integrates theoretical knowledge and practical configuration skills related to:

- Network segmentation
- Device hardening
- Access Control Lists (ACLs)
- Firewall configuration
- Virtual Private Networks (VPNs)
- Threat mitigation techniques

Students will simulate a real-world business environment and apply appropriate security controls following industry best practices.

2. Assignment Scenario

A company, **ABC Tech Solutions**, is establishing a new branch office and requires a secure and scalable network infrastructure. The company has the following requirements:

- Segmented internal network
- Secure internet connectivity
- Protection against internal and external threats
- Secure remote access for employees
- Controlled access to network resources

Your group is appointed as the Network Security Team responsible for designing and implementing a secure solution.

3. Learning Outcomes Assessed

Upon successful completion of this assignment, students will be able to:

1. Design a secure network topology using Cisco devices.
2. Configure VLANs and implement inter-VLAN routing securely.

3. Apply device hardening techniques.
 4. Implement and verify Access Control Lists (ACLs).
 5. Configure firewall and NAT policies.
 6. Implement VPN solutions.
 7. Analyze common network attacks and recommend mitigation strategies.
 8. Produce professional technical documentation.
-

4. Group Requirements

- Group Size: 3–6 students
- Simulation Tools:
 - GNS3 / EVE-NG

Each group must clearly define member responsibilities in the report.

5. Technical Requirements

5.1 Network Design (20%)

Design a logical network topology that includes:

- Minimum 1 Router
- Minimum 2 Switches
- Firewall (Cisco ASA or router-based firewall or Other Firewall device like Fortinet, Palo Alto...)
- 1 Server (Web/DNS)
- At least 1 PC for each VLAN

Required VLANs:

VLAN ID	Name
10	Management
20	Staff
30	Guest
99	Native

Students must provide:

- Logical topology diagram
 - IP addressing scheme
 - VLAN allocation table
-

5.2 Device Hardening (15%)

Configure security best practices:

- Hostname configuration
- Secure passwords and enable secret
- Password encryption
- Login banner warning
- Disable unused ports
- Enable SSH and disable Telnet

Students must explain:

- Why SSH is preferred over Telnet
 - Importance of disabling unused ports
-

5.3 VLAN & Inter-VLAN Security (15%)

Students must:

- Configure VLANs
- Configure trunk links securely
- Implement Inter-VLAN routing
- Mitigate VLAN hopping attacks

Students must include explanation of:

- VLAN hopping attack
 - Prevention techniques
-

5.4 Access Control Lists (ACLs) (20%)

Implement:

- Standard ACL
- Extended ACL

Security Requirements:

- Guest VLAN must not access Management VLAN
- Staff VLAN may access internal server
- Only Management VLAN may SSH to router

Include:

- ACL design logic
 - Configuration screenshots
 - Testing results
-

5.5 Firewall and NAT Configuration (10%)

Configure:

- Inside, Outside, and (if applicable) DMZ interfaces
- NAT or PAT
- Allow public access to web server
- Block unauthorized inbound traffic

Explain:

- Difference between Stateful and Stateless firewall
-

5.6 VPN Implementation (10%)

Configure either:

- Site-to-Site VPN (advanced), and
- Remote Access VPN

Students must explain:

- IPsec concept
 - Difference between Site-to-Site and Remote Access VPN
-

5.7 Network Threat Analysis (10%)

Research and analyze the following attacks:

- Denial of Service (DoS)
- ARP Spoofing
- MAC Flooding
- STP Attack

Students must:

- Explain how each attack works
 - Describe mitigation techniques using Cisco features
 - Relate threats to their own topology
-

6. Deliverables

Each group must submit:

1. Simulation project file (include IOS of each network device)
2. Network topology diagram
3. Technical Report (15–25 pages), including:
 - Cover page
 - Introduction
 - Network design explanation
 - Configuration details

- Security justification
- Testing and verification results
- Threat analysis
- Conclusion
- References (APA format)

4. Presentation slides (15 slides)

7. Presentation Requirements

- Duration: 15 minutes per group
 - Each member must participate
 - Live demonstration required
 - Q&A session (5 minutes)
-

8. Marking Scheme

Criteria	Weight
Network Design	20%
Configuration Accuracy	25%
Security Implementation	25%
Documentation Quality	15%
Presentation & Demonstration	15%
Video record	

9. Academic Integrity

Students must ensure:

- All configurations are developed by the group
- Proper citation of external sources
- No plagiarism
- No sharing of configuration files between groups

Any academic misconduct will be handled according to institutional regulations.

10. Bonus (Optional – Up to 5%)

Additional security implementations may earn bonus marks:

- Port Security
- DHCP Snooping
- Dynamic ARP Inspection

- Syslog configuration
 - Secure NTP configuration
 - SNMPv3
-

11. Submission Deadline

Submission Date: 19-Mar-2026

Submission Method: LMS

Late submissions may incur penalties according to university policy.

Scoring Rubric:

Total Assignment Weight: 100%

1. Network Design – 20%

Criteria	Marks
Logical topology completeness (router, switches, firewall, server, VLANs)	5%
Correct IP addressing scheme & subnetting	5%
VLAN design accuracy & segmentation logic	5%
Professional network diagram & documentation clarity	5%
Total	20%

High Distinction Expectation:

- Clean, scalable design
- Proper subnet planning
- Clear segmentation rationale
- Professionally drawn diagram

2. Device Hardening – 15%

Criteria	Marks
Secure password & enable secret configuration	3%
SSH configuration & Telnet disabled	4%
Banner, password encryption, hostname	3%
Unused ports disabled & documented	3%
Explanation of security importance	2%
Total	15%

High Distinction Expectation:

- Proper key generation for SSH
- Strong password policy
- Clear explanation of security reasoning

3. VLAN & Inter-VLAN Security – 15%

Criteria	Marks
Correct VLAN creation	4%
Trunk configuration (secure trunking)	4%

Inter-VLAN routing working correctly	4%
VLAN hopping mitigation explanation	3%
Total	15%

High Distinction Expectation:

- Native VLAN changed
- DTP disabled
- Access ports manually configured
- Clear explanation of VLAN hopping mitigation

4. Access Control Lists (ACLs) – 20%

Criteria	Marks
Correct standard ACL implementation	5%
Correct extended ACL implementation	5%
ACL logic meets scenario requirements	5%
Testing verification & screenshots	3%
Explanation of ACL placement & logic	2%
Total	20%

High Distinction Expectation:

- Correct inbound/outbound placement
- Logical sequencing
- No unnecessary rules
- Evidence of testing (ping, SSH tests)

5. Firewall & NAT Configuration – 10%

Criteria	Marks
Inside/Outside/DMZ configuration	3%
NAT/PAT correctly configured	3%
Web server access from outside properly secured	2%
Explanation of Stateful vs Stateless firewall	2%
Total	10%

High Distinction Expectation:

- Controlled inbound rules

- Minimal open ports
- Clear firewall policy logic

6. VPN Implementation – 10%

Criteria	Marks
Correct VPN configuration	4%
Connectivity verification	3%
IPsec explanation	2%
Comparison of Site-to-Site vs Remote Access	1%
Total	10%

High Distinction Expectation:

- Working encrypted tunnel
- Proper authentication configuration
- Clear explanation of Phase 1 & Phase 2 (if advanced)

7. Threat Analysis – 10%

Criteria	Marks
Technical explanation of attacks	3%
Application to own topology	3%
Cisco-based mitigation techniques	3%
Risk evaluation & clarity	1%
Total	10%

High Distinction Expectation:

- Clear attack flow explanation
- Identification of vulnerable VLAN/device
- Correct mitigation features (DAI, Port Security, BPDU Guard)
- Simple risk assessment table

8. Documentation & Report Quality – 5%

Criteria	Marks
Professional formatting	2%
Clear structure & logical flow	1%

Screenshots properly labeled	1%
References & citation (APA)	1%
Total	5%

9. Presentation & Demonstration – 5%

Criteria	Marks
Technical clarity	2%
Equal participation	1%
Live demo functionality	1%
Ability to answer questions	1%
Total	5%

Overall Mark Distribution Summary

Section	Weight
Network Design	20%
Device Hardening	15%
VLAN & Inter-VLAN	15%
ACLs	20%
Firewall & NAT	10%
VPN	10%
Threat Analysis	10%
Documentation	5%
Presentation	5%
Total	100%